

# EDGE-AI BASED VIBRATION MONITORING AND ANAMOLY DETECTION DEVICE

**Muruganandam M**

Electrical and Electronics  
Engineering Panimalar  
Engineering College  
Chennai, India

**ManojKumar P**

Electrical and Electronics  
Engineering Panimalar  
Engineering College  
Chennai, India

**Malan S**

Electrical and Electronics  
Engineering Panimalar  
Engineering College  
Chennai, India

**MR. Padmarasan M, M.E**

Assistant Professor  
Electrical and Electronics Engineering  
Panimalar Engineering College Chennai, India

**Wahith S**

Electrical and Electronics Engineering  
Panimalar Engineering College Chennai, India

**Abstract—** This study presents a universal Edge-AI powered vibration monitoring device for real-time anomaly detection across various applications. The MPU6050 sensor is used by the device to gather acceleration and gyroscope data, which is then locally processed on the ESP32 with Cat Boost to detect anomalies. Kurtosis, RMS, and variance are among the real-time features that the system extracts by utilizing edge computing, guaranteeing quick and effective anomaly identification without the need for preset thresholds. When anomalies are found, immediate notifications are sent by email, or WhatsApp, and a Thing Speak dashboard shows data in real time for remote monitoring in a graphical manner. This adaptable solution can be used in motion-sensitive environments such as transportation systems, industrial machinery, and structural health monitoring, improving system reliability and predictive maintenance.

## I. INTRODUCTION

Traditional methods for monitoring vibration often fall short in delivering real-time insights, primarily due to their dependence on predefined thresholds and cloud-based processing, which can result in delays in identifying faults. To address these challenges, this project introduces an edge- AI powered system that employs the MPU6050 sensor and ESP32 for real-time motion analysis and anomaly detection. The system locally processes acceleration and gyroscope data using machine learning techniques to detect deviations from standard behavior without the need for continuous cloud connectivity. By leveraging edge computing, essential features such as kurtosis, RMS, and variance are extracted and analyzed promptly, facilitating swift identification of irregular patterns. Upon detecting an anomaly, the system promptly issues alerts through WhatsApp, email, while simultaneously transmitting relevant data to a Thing Speak dashboard for visualization. This innovative solution improves predictive maintenance, minimizes downtime, and provides a versatile approach applicable to industrial machinery, structural health monitoring, transportation systems, and other motion-sensitive applications.

## II. SYSTEM ANALYSIS

### A. EXISTING SYSTEM:

Conventional vibration monitoring systems often rely on fixed thresholds, manual assessments, or cloud-based processing, which frequently fail to provide accurate real- time anomaly detection. Many of these conventional systems employ rudimentary statistical techniques or fixed threshold limits to identify unusual behavior, rendering them less flexible in adapting to intricate patterns and environmental changes. Consequently, these systems frequently encounter issues with false alarms and missed detections, undermining their reliability for predictive maintenance.

Monitoring solutions that rely on cloud infrastructure introduce delays, as raw sensor data must be sent to remote servers for analysis before actionable insights can be obtained. This method not only prolongs response times but also demands considerable bandwidth and energy, making it unsuitable for real-time applications. Moreover, dependence on centralized processing restricts the scalability and implementation of such systems in areas with unreliable or sporadic network connectivity.

Another significant drawback of current systems is their limited capacity to effectively differentiate between various types of anomalies. Many traditional solutions do not leverage advanced machine learning algorithms, which hampers their ability to adapt and learn from actual operational conditions. Furthermore, most systems lack a built-in communication mechanism for immediate alerts, necessitating that users manually monitor dashboards or logs to identify irregularities, which can result in delayed reactions.

Given these limitations, there is a pressing need for a sophisticated, real-time anomaly detection system powered by edge AI that can process vibration data locally. By incorporating machine learning on an edge device such as the ESP32 and utilizing real-time communication methods like WhatsApp, email for anomaly notifications, the proposed system seeks to offer a more efficient, scalable, and precise monitoring solution applicable to a wide range of fields, including industrial machinery, structural health monitoring, and transportation systems.

### B. Our Project Explanation:

The proposed system aims to deliver a thorough, real-time solution for anomaly detection monitoring by integrating the



Figure-2 Hardware setup

MPU6050 sensor, ESP32 microcontroller, Edge AI, and IoT communication technologies. It gathers vibration data from the MPU6050 and processes this information locally using machine learning algorithms on the ESP32, thereby leveraging edge computing for swift anomaly detection without dependence on cloud processing. Upon detecting an anomaly, the system promptly issues alerts through WhatsApp, email to inform the appropriate personnel.

Furthermore, all sensor data is relayed to an IoT dashboard, such as Thing Speak, where it is visualized through real-time graphs, widgets, and other visual indicators.

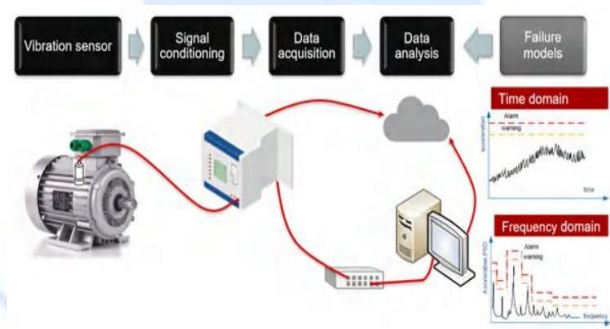


Figure-1 Workflow Diagram

The system also features local notifications to ensure immediate awareness on-site. A LED indicator provide visual signals whenever a fault or anomaly is identified, ensuring that both on-site staff and remote monitoring teams receive timely alerts. By integrating edge computing, real-time monitoring, and advanced anomaly detection, the system reduces downtime, enhances reliability, and supports predictive maintenance. Its scalable and cost-effective architecture makes it suitable for a wide range of applications, including monitoring industrial machinery, assessing structural health, diagnosing vehicle issues, and overseeing smart infrastructure.

#### 1. MPU6050 (Accelerometer & Gyroscope)

C.

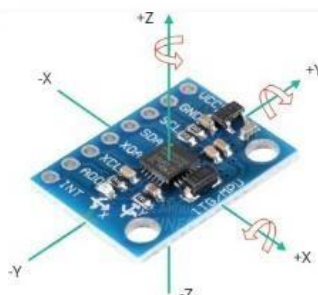


Figure-3 MPU6050 sensor module

The MPU6050 is a sophisticated 6-axis motion sensor that combines a 3-axis accelerometer with a 3-axis gyroscope, facilitating the measurement of motion and vibration in real-time. The accelerometer is responsible for detecting linear acceleration, whereas the gyroscope measures angular velocity, allowing for a thorough analysis of motion. In this project, the MPU6050 captures data related to vibration and tilt from the machine under observation. This data is instrumental in identifying anomalies such as misalignment, imbalance, and excessive vibrations, which may signal potential faults or wear in the machinery. Its high sensitivity and capability for real-time data acquisition ensure accurate monitoring, enabling early detection of faults.

## 1. ESP32 (Microcontroller &amp; Edge Computing Unit)



Figure-4 ESP32 MCU

The ESP32 serves as the system's core processing unit, overseeing the real-time collection and analysis of data from the MPU6050. It locally processes vibration and tilt information using Edge AI, which allows for immediate anomaly detection without depending on cloud services. The ESP32 is equipped with Wi-Fi and MQTT capabilities,

facilitating smooth data transmission to the dashboard while also promoting low-power operation for efficient battery consumption. By conducting FFT (Fast Fourier Transform) analysis on the sensor data, the ESP32 identifies significant vibration frequency patterns, aiding in the detection of abnormal conditions in both industrial and general monitoring scenarios.

## 2. Battery Backup (Power Supply Unit)

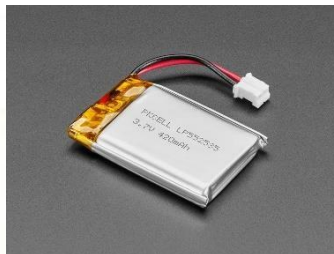


Figure-5 Lithium-ion rechargeable battery

The system operates on a 3.7V 1500mAh rechargeable battery, which facilitates both portability and continuous functionality. Due to the demands of real-time data acquisition and edge computing, power efficiency is a critical factor. To reduce energy consumption during periods of inactivity, the ESP32 employs a deep sleep mode, which significantly prolongs battery life. Furthermore, low-power enhancements are implemented for the MPU6050 sensor, which decreases power usage while still providing precise monitoring of vibration. The rechargeable battery supports extended use and can be paired with power management circuits to control voltage and guarantee the stable operation of all system components.

## 4 Neo pixel (Status indication)



Figure-6 Neo pixel

The Neo Pixel LED is used for real-time visual indication of system status and fault detection. Unlike traditional LEDs, Neo Pixel LEDs offer programmable multi-color output, allowing different colors to represent various conditions, such as normal operation, fault detection, or system errors. The microcontroller controls the Neo Pixel via a single data line, making it efficient and easy to integrate into the system. By providing clear and immediate visual feedback, the Neo Pixel enhances the usability of the fault detection system, ensuring quick recognition of issues by operators and maintenance personnel. *Schematic Diagram:*

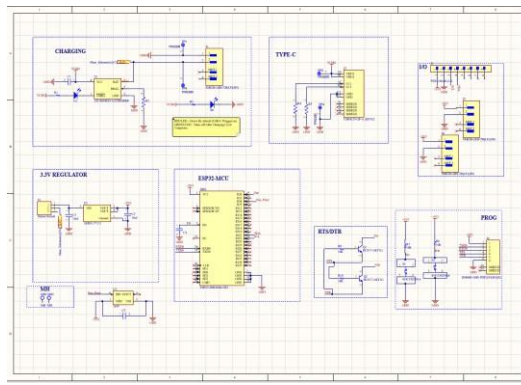


Figure-7 Schematic Diagram

#### D. Software used:

1. Embedded Programming (ESP32 & Edge Computing)



Figure-8 Arduino IDE

The ESP32 microcontroller is programmed through the Arduino IDE, taking advantage of its compatibility with the MPU6050 sensor and various wireless communication protocols. This system is engineered for real-time data processing, utilizing Edge AI methodologies to analyze information locally prior to transmission. Algorithms designed for power efficiency enhance computational performance while reducing latency, thereby facilitating seamless operation in battery-operated environments.

2. Sensor Data Acquisition & Preprocessing

The MPU6050 sensor gathers unprocessed data from accelerometers and gyroscopes, which is subsequently filtered and refined to eliminate noise and identify significant vibration patterns. Preprocessing methods, including moving averages, Fast Fourier Transform (FFT), and feature extraction techniques such as kurtosis, are employed before the data is input into the anomaly detection model.

3. Machine Learning-Based Anomaly Detection



Figure-9 Edge Impulse

The system utilizes Cat Boost, Support Vector Machines (SVM), machine learning algorithms to identify anomalies within vibration data. Initially, the device establishes a baseline of normal vibration patterns for the monitored machinery. During ongoing monitoring, the incoming data is assessed against this established model, with any discrepancies classified as anomalies. This methodology removes the reliance on fixed thresholds, allowing the system to be both adaptive and precise.

4. Wireless Communication & Cloud Integration

Upon detecting an anomaly, the ESP32 sends alerts using the MQTT protocol to a Thing Speak dashboard, where real-time data is represented through graphs, widgets, and live status indicators. Notifications are also dispatched via email, WhatsApp, to inform relevant personnel. The system is designed to minimize data transmission by only relaying processed and pertinent information, thereby optimizing both bandwidth usage and power efficiency.

## 5. User Dashboard and Visualization

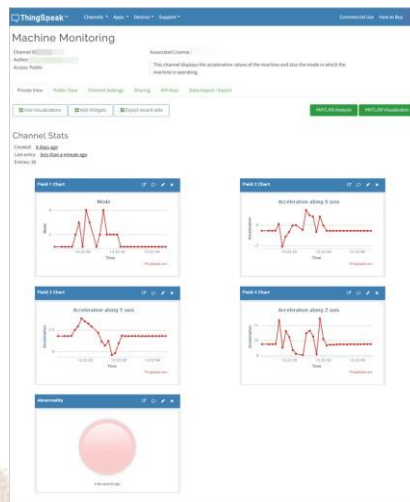


Figure-10 Think speak Dashboard

A tailored dashboard is created utilizing Thing Speak, offering real-time insights into machine vibrations. Users have access to historical trends, live sensor data, and alerts for anomalies presented in an organized manner. This dashboard facilitates remote monitoring, thereby improving predictive maintenance functions.

#### E. Results:

The proposed Edge-AI Based Intelligent Vibration Monitoring System proficiently identifies and categorizes anomalies in real-time, thereby ensuring dependable fault detection and remote oversight. The system underwent evaluation across a range of operational conditions, encompassing both standard vibration patterns and simulated fault scenarios, to determine its precision and effectiveness. Utilizing the MPU6050 sensor, the system successfully gathered acceleration and gyroscopic data, which was subsequently preprocessed and analyzed through edge computing methodologies on the ESP32 platform. Sophisticated machine learning algorithms, such as Cat Boost, SVM, and Light GBM, were implemented on the edge device to classify vibration patterns and identify deviations from expected operational behavior. The findings indicated a high level of accuracy in anomaly detection while maintaining minimal computational latency.

Integration with Thing Speak facilitated real-time visualization of sensor data via interactive dashboards, which featured time-series graphs, spectral analysis, and status indicators. The anomaly detection system effectively generated alerts through MQTT-based cloud notifications, providing immediate updates via email and WhatsApp. This strategy significantly enhanced the response time for maintenance teams, thereby minimizing potential downtime. Moreover, the system, powered by a 3.7V 1500mAh battery, exhibited energy-efficient performance, enabling prolonged continuous operation. The adoption of edge computing reduced the need for extensive data transmission to the cloud, optimizing bandwidth utilization and improving system responsiveness. The results affirm that the system is scalable, cost-effective, and versatile for a variety of practical applications, including industrial machine monitoring, predictive maintenance, automotive diagnostics, and structural health assessment.

### III. SOFTWARE OUTPUT

The software developed in this project facilitates real-time monitoring of vibrations by combining sensor data acquisition, edge AI-driven anomaly detection, cloud-based visualization, and automated alert systems. The MPU6050 accelerometer and gyroscope continuously gather motion data, which is processed locally by the ESP32 through I2C communication. Initially, the raw data, encompassing acceleration ( $A_x$ ,  $A_y$ ,  $A_z$ ) and gyroscope ( $G_x$ ,  $G_y$ ,  $G_z$ ) measurements, undergoes filtering and preprocessing to eliminate noise. Machine learning algorithms, including Cat Boost, SVM, and Light GBM, are employed on the ESP32 to analyze vibration patterns and identify anomalies, moving beyond conventional threshold-based detection methods.

Upon detecting an anomaly, the system promptly activates an alert through an LED flash and buzzer notification, ensuring immediate response.

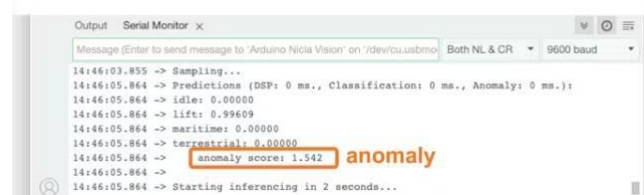


Figure-10 Serial monitor output

Subsequently, the processed data is sent via MQTT to a Thing speak dashboard, where graphical elements such as line charts and anomaly indicators offer real-time visualization of vibration trends. Furthermore, notifications are dispatched to the maintenance team through email and WhatsApp, providing comprehensive details about the identified fault, including sensor readings and timestamps. The system also tracks battery voltage, displaying it on the dashboard and issuing a low-battery alert when necessary. This comprehensive approach guarantees precise real-time monitoring, swift fault detection, and reduced downtime, rendering the system applicable across various sectors, including industrial, automotive, and structural health monitoring.

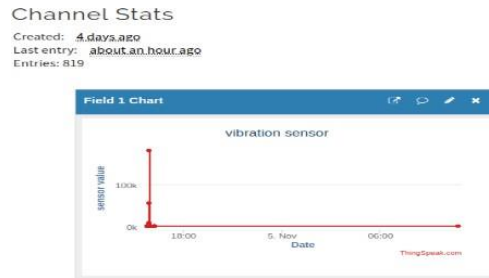


Figure-11 Graphical Representation

#### IV. CONCLUSION

In summary, the Edge-AI Based Intelligent Vibration Monitoring System addresses the shortcomings of conventional vibration monitoring techniques by utilizing MPU6050, ESP32, and edge computing to facilitate real-time anomaly detection. This system enables swift data processing through machine learning models, thereby removing the reliance on cloud-based analysis, and minimizing latency. Furthermore, the incorporation of Thing speak for data visualization, along with notifications via email and WhatsApp, enhances the system's responsiveness to identified anomalies. The findings indicate that this method significantly enhances real-time fault detection, boosts system reliability, and improves operational efficiency, positioning it as a cost-effective and scalable solution suitable for a range of industrial and portable applications.

#### Reference:

1. Lee, S.B.; Stone, G.C.; Antonino-Daviu, J.; Gyftakis, K.N.; Strangas, E.G.; Maussion, P.; Platero, C.A. Condition monitoring of industrial electric machines: State of the art and future challenges. *IEEE Ind. Electron. Mag.* 2020, 14, 158–167.
2. Madhukar, G.; Boosi, S.; Adhoni, Z.A.; Bhaskar, K.A.; Naik, A.V. A Machine Learning Based Methodology for Fault Diagnosis in Rotating Machine. In *Proceedings of the 2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS)*, Raichur, India, 24–25 February 2023; pp. 1–5.
3. Shukla, K.; Nefti-Meziani, S.; Davis, S. A heuristic approach on predictive maintenance techniques: Limitations and scope. *Adv. Mech. Eng.* 2022, 14, 16878132221101009.
4. Kumar, A.; Sathujoda, P.; Bhalla, N.A. Vibration signal analysis of a rotor-bearing system through wavelet transform and empirical mode decomposition. *IOP Conf. Ser. Mater. Sci. Eng.* 2022, 1248, 012027.
5. Efeoglu, E.; Tuna, G. Machine Learning for Predictive Maintenance: Support Vector Machines and Different Kernel Functions. *J. Mach. Manuf. Reliab.* 2022, 51, 447–456.
6. Villarroel, A.; Zurita, G.; Velarde, R. Development of a Low-Cost Vibration Measurement System for Industrial Applications. *Machines* 2019, 7, 12.
7. Umbrajkar, A.; Krishnamoorthy, A.; Dhumale, R. Vibration analysis of shaft misalignment using machine learning approach under variable load conditions. *Shock Vib.* 2020, 2020, 1650270.
8. Nascimento, E.G.S.; Liang, J.S.; Figueiredo, I.S.; Guarieiro, L.L.N. T4PdM: A Deep Neural Network based on the Transformer Architecture for Fault Diagnosis of Rotating Machinery. *arXiv* 2022, arXiv:2204.03725.
9. Althubaiti, A.; Elasha, F.; Teixeira, J.A. Fault diagnosis and health management of bearings in rotating equipment based on vibration analysis—A review. *J. Viborg.* 2022, 24, 46–74.
10. Tiboni, M.; Remino, C.; Bussola, R.; Amici, C. A review on vibration-based condition monitoring of rotating machinery. *Appl. Sci.* 2022, 12, 972.
11. Ranjan, R.; Ghosh, S.K.; Kumar, M. Fault diagnosis of journal bearing in a hydropower plant using wear debris, vibration and temperature analysis: A case study. *Proc. Inst. Mech. Eng. Part E J. Process Mech. Eng.* 2020, 234, 235–242.
12. Li, Q.; Zhang, W.; Liu, J.; Chen, M. Fault diagnosis of wind turbine gearbox based on improved ELM algorithm and multimodal sensor fusion. *Measurement* 2021, 180, 109598.
13. Cheng, Y.; Lin, M.; Wu, J.; Zhu, H.; Shao, X. Intelligent fault diagnosis of rotating machinery based on continuous wavelet transform-local binary convolutional neural network. *Knowl.-Based Syst.* 2021, 216, 106796.